CSE276 - Artificial Intelligence Foundations | Section K3P25LH | CA-1 Groups and Projects

Find your group by your name or registration number. The objectives are what your project will be assessed on; the advanced extension is optional. Use the official source to read the original problem statement (for Smart India Hackathon problems, search the PS ID on the linked page).

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G1	Chayan Khatua (12518616) Aditya Kumar Singh (12509260)	**Legacy Land Record Digitisation and Validation Engine**	A large share of Indian land records exist only as handwritten registers and scanned legacy files, so ownership questions take months to answer.	Revenue departments, sub-registrar offices, landowners	1. Build a pipeline that reads scanned records, extracts structured fields, and flags entries that contradict the existing register for human review. 2. Produce a field extractor with per-field accuracy plus a contradiction report over a sample register. 3. Report per-field extraction accuracy, contradiction precision, share of records needing review. Baseline: template-based rule extraction. Deliver Streamlit reviewer: scanned page beside extracted fields.	OCR and handwriting recognition; information extraction; rule-based consistency checking	Unit II image preprocessing; Unit III-IV CNN; Unit I knowledge representation and rule checking	IAM Handwriting and Devanagari datasets (public), plus sample public land records (Find and cite suitable public data yourselves.)	Add confidence-based routing so only low-confidence fields reach a human.	SMART INDIA HACKATHON 2026 PS ID: SIH26018 Intelligent Land Record Digitization and Validation System Organisation: Ministry of Rural Development https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G2	Aman Kumar Aditya (12517923) Divam Sharma (12516865)	**Land Acquisition Delay Early-Warning System**	Infrastructure land acquisition slips by months and the delay is recognised only once the project has already stalled.	Project authorities, district administrations, contractors	1. Build a predictive system that scores each acquisition case for delay risk and names the specific stage causing it. 2. Produce a delay-risk model with a ranked watchlist and the driving stage per case. 3. Report rOC-AUC, precision at top-K watchlist, lead time before the delay materialises. Baseline: stage-average heuristic. Deliver Dashboard with a ranked case watchlist.	Classification on process data; feature importance; survival-style time-to-event framing	Unit III classification and trees; Unit II feature engineering; Unit V evaluation	Public infrastructure project monitoring data (e.g. MoSPI project monitoring reports) (Find and cite suitable public data yourselves.)	Model time-to-completion rather than a binary delay flag.	SMART INDIA HACKATHON 2026 PS ID: SIH26017 Predictive Analytics System for Early Detection of Land Acquisition Delays Organisation: Ministry of Rural Development https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G3	Ankit Sharma (12513795) Krishna (12515666)	**Public Procurement Bid Compliance Verification Assistant**	Procurement officers manually verify statutory, tax and eligibility documents for every bid, which is slow and where costly mistakes happen.	Government procurement officers, bidders, audit teams	1. Build an assistant that checks each bid against encoded eligibility rules, states which rule failed, and escalates ambiguity. 2. Produce a compliance checker over sample bids with per-rule accuracy and an escalation rate. 3. Report per-rule precision/recall, escalation recall, share auto-cleared. Baseline: keyword checklist. Deliver Streamlit checker: upload bid, get pass/fail with reasons.	Document field extraction; rule-based eligibility reasoning; abstention and escalation	Unit I knowledge representation and rule-based reasoning; Unit V NLP; Unit V evaluation	Public GeM procurement rules and published tender documents (Find and cite suitable public data yourselves.)	Add a cost model weighing a missed escalation against a false one.	SMART INDIA HACKATHON 2026 PS ID: SIH26100 AI-Powered Integrated Bid Compliance Verification Platform for GeM Procurement Organisation: Ministry of Petroleum & Natural Gas https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G4	Aman Sharma (12511704) Ayush Raj (12516654)	**Indian Standards Advisory Assistant for Industry**	Manufacturers and consumers cannot find which Indian Standard applies to them, so compliance is guessed or outsourced.	MSMEs, consumers, BIS enquiry desks	1. Build a grounded question-answering assistant over the standards corpus that cites the exact clause and refuses when unsupported. 2. Produce a cited-answer assistant evaluated on 100 questions with known answers. 3. Report answer correctness and citation correctness reported separately; refusal rate. Baseline: keyword search over the corpus. Deliver Streamlit assistant with clause citations.	Retrieval-augmented generation; semantic search; abstention	Unit V NLP and text representation; Unit I knowledge representation; Unit V evaluation	Published BIS standards documents (public) (Find and cite suitable public data yourselves.)	Add multilingual query handling for regional-language enquiries.	SMART INDIA HACKATHON 2026 PS ID: SIH26107 AI-powered Intelligent Assistant for Indian Standards and BIS Services for Industries and Consumers Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G5	Himanshu Singh (12512235) Aman Katoch (12519801)	**Procurement Specification to Standards Recommendation Engine**	Identifying the correct Indian Standards for a procurement specification is hard because scopes overlap and revisions are frequent.	Procurement agencies, PSU technical committees	1. Build a recommender that maps a free-text specification to the applicable standards and explains each recommendation. 2. Produce a ranked standards recommender evaluated against expert-labelled specifications. 3. Report recall@K, MAP, NDCG. Baseline: TF-IDF keyword match. Deliver Streamlit tool: specification in, ranked standards out.	Semantic similarity and ranking; multi-label recommendation; explanation of matches	Unit V NLP and embeddings; Unit III classification; Unit V ranking metrics	Published BIS standards catalogue (public) (Find and cite suitable public data yourselves.)	Handle superseded standards so revisions do not silently mislead.	SMART INDIA HACKATHON 2026 PS ID: SIH26108 AI-Powered Recommendation Engine for Identifying Applicable Indian Standards for Procurement Specifications Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G6	Sabuj Mondal (12504433) Sindhant Dhadwal (12509119)	**Material Code Harmonisation Across Public Enterprises**	The same material carries different codes and descriptions across enterprises, causing duplicate stock and fragmented procurement data.	Public sector enterprises, inventory and procurement teams	1. Build a matching system that groups functionally equivalent material records and proposes one harmonised code with evidence. 2. Produce an entity resolution model with pairwise precision/recall over a labelled sample. 3. Report pairwise precision/recall/F1, cluster purity. Baseline: exact string match. Deliver Reviewer UI showing proposed merges with evidence.	Text similarity and entity resolution; clustering of near-duplicates; large-label classification	Unit V NLP; Unit VI clustering; Unit V evaluation of matching	Public material master and procurement catalogues (Find and cite suitable public data yourselves.)	Add active learning so a reviewer resolves only the ambiguous pairs.	SMART INDIA HACKATHON 2026 PS ID: SIH26099 AI-Driven Standardization and Harmonization of Material Codes Across CPSEs Organisation: Ministry of Petroleum & Natural Gas https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G7	Sankalp Jeet Singh (12512484) Laksh Mahajan (12512521)	**Public Scheme Expenditure Anomaly and Fraud Screening**	Scheme expenditure is published but rarely analysed, so irregularities surface only through audit years later.	Audit bodies, scheme administrators, citizens	1. Build a screening system that ranks expenditure records by anomaly strength for auditor review, with the reason attached. 2. Produce a ranked anomaly list with manual verification of a sample and reported precision. 3. Report precision at top-K on manual verification, flag rate. Baseline: simple amount-threshold rule. Deliver Auditor dashboard with ranked flags.	Unsupervised anomaly detection; feature engineering on spending records; explanation of flags	Unit VI anomaly detection; Unit II feature engineering; Unit V rare-event evaluation	Open Government Data India expenditure datasets (public) (Find and cite suitable public data yourselves.)	Add network features linking vendors across works.	SMART INDIA HACKATHON 2026 PS ID: SIH26102 Development of an AI-powered system to detect anomalies, fraud, and inefficiencies in MPLAD Scheme implementation regd. Organisation: MoSPI https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G8	Shashi Ranjan (12514670) Pradyuman Chandel (12513411)	**Government Scheme Eligibility Matching with Justified Decisions**	Applicants miss support they qualify for because scheme rules are scattered and eligibility is never explained.	Marginalised applicants, scheme administrators, intermediary agencies	1. Build an advisor that determines eligibility across schemes and states exactly which condition decided each outcome. 2. Produce an eligibility engine validated against manually determined outcomes on test profiles. 3. Report decision accuracy against ground truth, explanation completeness. Baseline: single-scheme checklist. Deliver Streamlit advisor: profile in, eligible schemes with reasons.	Rule-based reasoning over encoded scheme criteria; explanation trace	Unit I knowledge representation, rules, defaults and exceptions; Unit V evaluation	Published scheme rules from myScheme and state portals (Find and cite suitable public data yourselves.)	Add a cost-aware ranking of schemes by expected benefit.	SMART INDIA HACKATHON 2026 PS ID: SIH26092 AI-Driven Scheme Matching for Marginalized Entrepreneurs Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G9	Pogiri Harshavardhan (12519723) Pinnintla Haresh Reddy (12512073)	**Packaged Commodity Declaration Compliance Checker**	Every packaged commodity must carry mandatory declarations, and checking them across retail and e-commerce is done by hand.	Legal Metrology inspectors, e-commerce platforms, consumers	1. Build a checker that reads a package label image and verifies each mandatory declaration against the encoded rules. 2. Produce a label checker reporting per-declaration detection accuracy and overall compliance verdicts. 3. Report per-declaration precision/recall, verdict accuracy. Baseline: OCR plus keyword presence. Deliver Streamlit tool: label photo in, compliance report out.	OCR; text field extraction; rule-based compliance verification	Unit II image preprocessing; Unit III-IV CNN and OCR; Unit I rule reasoning	Public product label image datasets plus Legal Metrology rule text (Find and cite suitable public data yourselves.)	Handle multilingual and low-contrast labels.	SMART INDIA HACKATHON 2026 PS ID: SIH26034 Software System to check compliance of Packaged Commodities under Legal Metrology(Packaged Commodities) Rules, 2011 by scanning products, images and labels. Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS
G10	Manish Kumar (12515548) Pranav Kumar (12515373)	**Railway Maintenance Block Planning for Asset Availability**	Maintenance blocks are requested independently by departments, producing conflicting closures and lost train availability.	Railway operations and engineering departments	1. Build a planner that schedules maintenance blocks jointly across departments to maximise asset availability. 2. Produce a block scheduler compared against independent departmental scheduling on availability lost. 3. Report total availability, conflicts resolved, schedule feasibility. Baseline: first-come-first-served requests. Deliver Gantt-style scheduling dashboard.	Constraint satisfaction and scheduling; conflict detection; optimisation search	Unit I search, planning and CSP; Unit V evaluation of policies	Public railway timetable and maintenance window data (Find and cite suitable public data yourselves.)	Add stochastic overrun so plans are robust to jobs finishing late.	SMART INDIA HACKATHON 2026 PS ID: SIH26027 AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways Organisation: Ministry of Railways https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G11	Sulagna Bhattacharyya (12509165) Abhishek Rawat (12510193)	**Dynamic Train Arrival Time Forecasting**	Published arrival times drift badly during the journey, so passengers and connecting services plan on numbers that are already wrong.	Railway passengers, station operations, connecting transport	1. Build a forecasting service that updates expected arrival continuously and reports how far ahead it stays reliable. 2. Produce an ETA model with error reported by remaining distance, against a scheduled-time baseline. 3. Report mAE and RMSE by remaining distance; share within tolerance. Baseline: published schedule, mandatory. Deliver Live ETA dashboard per train.	Time-series regression; en-route feature engineering; error decay analysis	Unit VI time-series; Unit II feature engineering; Unit V forecast evaluation	Public Indian Railways timetable and running status datasets (Find and cite suitable public data yourselves.)	Propagate delay through the network to downstream stations.	SMART INDIA HACKATHON 2026 PS ID: SIH26028 Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains Organisation: Ministry of Railways https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G12	Garima Vaid (12516225) Anshu (12516312)	**Regional Logistics and Accessibility Routing Platform**	Difficult terrain and frequent disruption mean the nominal route to a remote district is often unusable, delaying essential goods.	State logistics departments, medical supply chains, remote districts	1. Build a routing platform that plans deliveries around disruption and reports which settlements remain unreachable. 2. Produce a router that reports reachability and delivery time under simulated disruptions. 3. Report reachability percentage, delivery time, unreachable settlement count. Baseline: shortest-distance routing. Deliver Map dashboard with routes and unreachable areas.	Graph search and constrained shortest path; disruption modelling	Unit I search and heuristic search; Unit II data preparation; Unit V evaluation	OpenStreetMap road network and public disruption reports (Find and cite suitable public data yourselves.)	Add multi-modal legs where road alone fails.	SMART INDIA HACKATHON 2026 PS ID: SIH26002 AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER) Organisation: Ministry of Development of North Eastern Region (MDoNER) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G13	Navdeep Singh (12518471) Dhanvir Singh (12520115)	**Metaheuristic Traffic Route Optimisation for Congested Corridors**	Route guidance optimises each vehicle separately, which pushes everyone onto the same corridor and creates the congestion it was avoiding.	City traffic authorities, navigation providers, commuters	1. Build a system-level router that assigns flows across corridors and reports total travel time saved against selfish routing. 2. Produce a flow assignment compared against per-vehicle shortest-path routing on total network time. 3. Report total network travel time, worst-corridor delay, convergence. Baseline: selfish shortest path, mandatory. Deliver Network map with before/after flow visualisation.	Metaheuristic optimisation; traffic assignment; comparative search strategies	Unit I search and optimisation; Unit V policy evaluation	OpenStreetMap network with public traffic count data (Find and cite suitable public data yourselves.)	Compare two metaheuristics under an equal evaluation budget.	SMART INDIA HACKATHON 2026 PS ID: SIH26137 Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization Organisation: Egreen Quanta https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G14	Inesh Pal Singh (12512728) Arshpreet Singh (12513377)	**Decentralised Warehouse Robot Fleet Coordination**	Central path planning for warehouse robots fails on network latency and dead zones, and deadlocks stop the whole floor.	Warehouse operators, fulfilment centres	1. Build a decentralised coordination policy where robots negotiate locally, evaluated on throughput and deadlock frequency. 2. Produce a coordination policy compared against centralised planning across fleet sizes. 3. Report throughput, deadlock rate, message count. Baseline: centralised planner and naive collision avoidance. Deliver Animated warehouse simulation with metrics panel.	Multi-agent pathfinding; local coordination protocols; simulation	Unit I search and multi-agent systems; Unit V policy evaluation	Open multi-agent pathfinding benchmarks and simulators	Add communication dropout and measure resilience.	SMART INDIA HACKATHON 2026 PS ID: SIH26123 Edge-AI Based Distributed Fleet Coordination for Autonomous Mobile Robots (AMRs) in Smart Warehouses Organisation: Bharat Electronics Limited https://sih.gov.in/sih2026PS
G15	Pushpak Kumar (12508875) Ritik Kumar (12502175)	**Modular Autonomous Mobile Robot Task Allocation**	A warehouse AMR platform is only as good as the policy deciding which robot takes which task, and that is usually first-come-first-served.	Warehouse automation integrators, operations teams	1. Build a task allocation policy for a heterogeneous AMR fleet, evaluated on completion time and fairness of robot utilisation. 2. Produce an allocation policy compared against nearest-robot assignment on makespan. 3. Report makespan, utilisation spread, tasks completed per hour. Baseline: nearest-available robot. Deliver Simulation dashboard with allocation timeline.	Assignment and auction-based allocation; multi-agent coordination	Unit I search, optimisation and agents; Unit V evaluation	Open warehouse simulation environments (Find and cite suitable public data yourselves.)	Handle robot failure mid-task and re-allocate.	SMART INDIA HACKATHON 2026 PS ID: SIH26112 Design and Develop a Modular Autonomous Mobile Robot (AMR) Platform for Smart Warehouse Automation Organisation: Autodesk https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G16	Umanshu Manan (12509488) Suryansh Thakur (12504629)	**Vision-Based Outdoor Navigation for Unmanned Ground Vehicles**	Outdoor ground vehicles lose GPS under canopy and in built-up areas, and must navigate from camera input alone.	Field robotics teams, agricultural and security operators	1. Build a vision-based navigation stack that follows a traversable path and reports failure conditions honestly. 2. Produce a navigation policy with success rate and a characterised failure mode list. 3. Report success rate, mean deviation from path, failures per run. Baseline: colour-threshold path following. Deliver Recorded run video plus a metrics report.	Semantic segmentation of traversable terrain; visual servoing; failure detection	Unit III-IV CNN and segmentation; Unit I search for local planning; Unit V evaluation	Public off-road driving and traversability datasets (Find and cite suitable public data yourselves.)	Add terrain-type awareness so the vehicle slows before rough ground.	SMART INDIA HACKATHON 2026 PS ID: SIH26126 Vision Based Autonomous Navigation for Unmanned Ground Vehicle for Outdoor environment Organisation: Bharat Electronics Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G17	Vajrala Chadeek (12504949) Syed Shahid (12509761)	**Adaptive Path Planning for Unstructured Indian Roads**	Autonomous driving stacks assume lane discipline and predictable behaviour, neither of which holds on Indian roads.	Autonomous vehicle developers, road safety researchers	1. Build a local planner that handles mixed traffic and sudden manoeuvres, evaluated on collision-free completion in simulation. 2. Produce a planner evaluated across traffic densities against a lane-following baseline. 3. Report collision-free completion rate, time to goal, minimum clearance. Baseline: rule-based lane following. Deliver Simulation replay with planner visualisation.	Local path planning and collision avoidance; behaviour prediction; simulation	Unit I search and planning; Unit III prediction; Unit V evaluation	India Driving Dataset (IDD) and open driving simulators (Find and cite suitable public data yourselves.)	Add a simple intent predictor for surrounding vehicles.	SMART INDIA HACKATHON 2026 PS ID: SIH26037 Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads Organisation: MathWorks https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G18	Gabu Manas Sri Charan (12501491) Shaik Neeraj (12513358)	**Low-Visibility Mine Vehicle Safety Assistant**	Haul vehicles in open-cast mines operate in fog and dust where operators cannot see obstacles until far too late.	Mine operators, haul vehicle drivers, safety officers	1. Build a perception assistant that detects vehicles and people in degraded visibility and issues a graded collision warning. 2. Produce a detector evaluated across visibility levels with a warning rule and false-alarm rate. 3. Report mAP by visibility band, warning lead time, false alarms per hour. Baseline: detector without enhancement. Deliver Operator display with proximity warnings.	Object detection under degraded imaging; image enhancement; alert thresholding	Unit II image preprocessing; Unit III-IV CNN detection; Unit V rare-event metrics	Public foggy/adverse-condition driving datasets (Find and cite suitable public data yourselves.)	Fuse an additional modality where cameras fail entirely.	SMART INDIA HACKATHON 2026 PS ID: SIH26007 Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines. Organisation: Ministry of Steel https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G19	Ashish Gupta (12509896) Abhay Kumar (12509464)	**Underground Mine Safety Monitoring and Rescue Support**	Underground incidents are discovered late and rescue teams enter without knowing conditions or worker locations.	Mine safety departments, rescue teams, underground workers	1. Build a monitoring system that fuses environmental sensing into a risk state and supports rescue prioritisation. 2. Produce a risk classifier with alert lead time against recorded incidents. 3. Report detection rate, lead time, false alarms per shift. Baseline: fixed gas-threshold alarms. Deliver Control-room dashboard with zone risk map.	Multivariate anomaly detection; risk state classification; alert logic	Unit VI anomaly detection; Unit III classification; Unit V rare-event evaluation	Public mine environmental monitoring datasets (Find and cite suitable public data yourselves.)	Add zone-level prioritisation for rescue routing.	SMART INDIA HACKATHON 2026 PS ID: SIH26039 AI-Powered Underground Mine Safety, Monitoring and Rescue System. Organisation: Government of Jharkhand https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G20	Mohd Umar Farooq Khan (12509289) Priyanshu Mishra (12516568)	**Coal Mine Governance and Compliance Monitoring System**	Statutory compliance across many mine sites is tracked on spreadsheets, so lapses are found at inspection rather than prevented.	Ministry of Coal, mine operators, inspectors	1. Build a monitoring system that scores each site for compliance risk and points to the specific overdue obligation. 2. Produce a compliance risk model with a ranked inspection list and the driving obligation. 3. Report precision at top-K inspections, obligation detection accuracy. Baseline: overdue-count ranking. Deliver Compliance dashboard with site risk ranking.	Rule-based obligation tracking; risk classification; document evidence extraction	Unit I rule reasoning; Unit III classification; Unit V evaluation	Published mine compliance and inspection records (Find and cite suitable public data yourselves.)	Add automatic evidence extraction from submitted reports.	SMART INDIA HACKATHON 2026 PS ID: SIH26024 AI-Based Smart Governance and Compliance Monitoring System for Coal Mines Organisation: Ministry of Coal https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G21	Nikit (12511334) Prince (12514381)	**Conveyor Belt Damage Prediction for Bulk Material Handling**	Conveyor belt joint failure stops an entire mining operation and is currently found only after rupture.	Mining operators, maintenance engineers	1. Build a prediction system that scores belt condition from monitoring signals and forecasts time to intervention. 2. Produce a condition model with warning lead time against recorded failures. 3. Report lead time, detection rate, false alarms per week. Baseline: fixed vibration threshold. Deliver Maintenance dashboard with belt health trend.	Time-series feature extraction; degradation trend detection; anomaly scoring	Unit VI time-series and anomaly detection; Unit II feature engineering; Unit V evaluation	NASA C-MAPSS and public industrial condition monitoring datasets (Find and cite suitable public data yourselves.)	Add remaining-useful-life estimation with uncertainty.	SMART INDIA HACKATHON 2026 PS ID: SIH26008 Belt Joint Rupture and Conveyor Belt Damages in Iron Ore Mining Industry: Intelligent Monitoring and Prediction of Conveyor Belt Joint Rupture and Damages in Iron Ore Mining Industry. Organisation: Ministry of Steel https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G22	Abhishek Nair (12508211) Rinshad C (12515201)	**Latent Defect Screening in Electronic Component Burn-In**	Burn-in screening uses static pass/fail limits, so components with latent defects pass and fail in the field.	High-reliability electronics manufacturers, space and defence integrators	1. Build a screening system that scores parametric drift during burn-in rather than checking a fixed limit. 2. Produce a drift-based screening model compared against static limits on escape rate. 3. Report escape rate, false reject rate, PR-AUC. Baseline: static parametric limits, mandatory. Deliver Screening dashboard with per-part drift traces.	Anomaly detection on parametric traces; drift feature engineering	Unit VI anomaly detection; Unit II feature engineering; Unit V imbalanced evaluation	SECOM manufacturing process dataset (public) (Find and cite suitable public data yourselves.)	Add per-batch baselines so normal drift is not flagged.	SMART INDIA HACKATHON 2026 PS ID: SIH26170 AI-Driven Anomaly Detection in Component Burn-In & Screening Organisation: Indian Space Research Organisation(ISRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G23	Ridham Sandle (12511031) Rohit R S (12510512)	**Objective Onion Quality Grading at Procurement Centres**	Onion grading is subjective and varies between procurement centres, so farmers are paid inconsistently for the same produce.	Procurement centres, farmers, agricultural marketing boards	1. Build a grading assistant that classifies quality from a photograph and produces a digital grade report instantly. 2. Produce a grader reporting per-defect accuracy and grade percentages against expert labels. 3. Report per-class precision/recall, grade percentage MAE, inter-centre variance. Baseline: colour-histogram classifier. Deliver Mobile-friendly grading app with instant report.	Image classification; defect detection; grade percentage estimation	Unit II image preprocessing; Unit III-IV CNN; Unit V multi-class metrics	Public produce quality image datasets (Find and cite suitable public data yourselves.)	Add per-centre calibration to remove systematic grader bias.	SMART INDIA HACKATHON 2026 PS ID: SIH26031 Quality assessment and grading of onions are often subjective and vary across procurement centers, resulting in disputes and inconsistencies. Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G24	Ananmay Saroch (12518907) Dhairya Jaiswal (12514042)	**Farmer Procurement Slot Scheduling and Wait Time Reduction**	Farmers queue for hours at procurement centres because arrivals are unmanaged and schedules are not published.	Procurement centres, farmers, mandi administrators	1. Build a scheduling system that allocates arrival slots and publishes expected wait, reducing total queueing. 2. Produce a slot allocation policy compared against unmanaged arrival on total wait. 3. Report mean and worst-case wait, throughput, fairness spread. Baseline: unmanaged first-come-first-served. Deliver Slot booking interface with live wait estimate.	Queue modelling; scheduling and assignment; demand forecasting	Unit I search and optimisation; Unit VI time-series for arrivals; Unit V evaluation	Public mandi arrival and procurement datasets (Find and cite suitable public data yourselves.)	Add fairness constraints so distant farmers are not disadvantaged.	SMART INDIA HACKATHON 2026 PS ID: SIH26032 Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status. Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G25	Jerom Jaice (12516484) Dhanush R (12514064)	**Forecast Bust Detection for Medium-Range Weather Prediction**	Medium-range forecasts occasionally fail badly during rapidly evolving systems, and operational users have no warning that today is such a day.	Meteorological services, disaster managers, agriculture advisories	1. Build a system that flags regions and lead times where the forecast is likely to be unreliable, so advisories can be hedged. 2. Produce a bust-risk classifier with precision on recorded bust events. 3. Report precision/recall on bust events, ROC-AUC, calibration. Baseline: ensemble spread threshold. Deliver Forecaster dashboard with reliability overlay.	Classification of forecast reliability; ensemble spread features; calibration	Unit II feature engineering; Unit III classification; Unit V calibration and rare-event metrics	Public NWP forecast archives and IMD observations (Find and cite suitable public data yourselves.)	Predict the magnitude of the error, not just its likelihood.	SMART INDIA HACKATHON 2026 PS ID: SIH26079 AI-Based Forecast Bust Detection for Medium-Range Weather Forecasts Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G26	Devarsh P N (12519387) Muhammed Sinan Aziz P P (12517501) Mrityunjay Chhetri (12519662)	**Regime-Aware Post-Processing of Monsoon Rainfall Forecasts**	Raw model rainfall forecasts carry systematic bias that differs between active and break monsoon phases, and uniform correction makes both worse.	Meteorological services, irrigation and agriculture planners	1. Build a post-processing system that identifies the monsoon regime first and applies the correction appropriate to it. 2. Produce a regime-aware corrector compared against uniform bias correction. 3. Report mAE and RMSE by regime, skill score against raw forecast. Baseline: uniform bias correction, mandatory. Deliver Dashboard comparing raw, uniform and regime-aware forecasts.	Regime classification then regression correction; conditional modelling	Unit VI clustering for regime discovery; Unit III regression; Unit V evaluation	IMD gridded rainfall and public NWP output (Find and cite suitable public data yourselves.)	Discover regimes unsupervised rather than using fixed definitions.	SMART INDIA HACKATHON 2026 PS ID: SIH26080 Regime-Aware AI Post-Processing of Monsoon Rainfall Forecasts Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G27	Pihu Agrawal (12518301) Gagan Jadaun (12505358)	**Spatio-Temporal Tracking of Extreme Weather Anomalies**	Extreme weather features are tracked manually across forecast frames, which does not scale and misses fast-moving systems.	Meteorological centres, disaster response agencies	1. Build a tracking system that identifies anomalous features and follows them across time, producing a movement forecast. 2. Produce a feature tracker with detection and track-continuity metrics against annotated events. 3. Report detection precision/recall, track fragmentation, position error. Baseline: per-frame threshold detection. Deliver Animated map showing tracked anomalies.	Anomaly detection in gridded fields; object tracking across frames	Unit VI anomaly detection; Unit III-IV CNN; Unit V detection metrics	Public reanalysis and gridded weather datasets (Find and cite suitable public data yourselves.)	Predict the next-step position of the tracked feature.	SMART INDIA HACKATHON 2026 PS ID: SIH26078 AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies in Medium-Range Forecasts Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G28	Krishika Mohindru (12502697) Utkarsh (12509410)	**Remote Research Station Operations Management Platform**	Polar and remote research stations are managed over intermittent links, so resource and maintenance decisions are made blind.	Station managers, expedition logistics, research agencies	1. Build a management platform that forecasts consumable depletion and flags equipment needing attention before resupply windows. 2. Produce a depletion forecaster and maintenance flag list evaluated over a simulated season. 3. Report forecast MAE, stock-out incidents avoided, flag precision. Baseline: linear extrapolation of usage. Deliver Station operations dashboard.	Consumption forecasting; anomaly detection on equipment telemetry	Unit VI time-series and anomaly detection; Unit II feature engineering; Unit V evaluation	Public facility consumption and equipment telemetry datasets (Find and cite suitable public data yourselves.)	Add resupply planning under a fixed window constraint.	SMART INDIA HACKATHON 2026 PS ID: SIH26060 Digital Platform for efficient remote management of Indian Antarctic Research Stations Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G29	Syed Mahammad Vahid (12519583) Ajay Kumar (12520098)	**Expedition Logistics and Asset Tracking Optimiser**	Expedition assets are tracked on spreadsheets, so equipment is shipped that already exists on site and critical items are missed.	Expedition planners, logistics coordinators	1. Build an optimiser that plans cargo manifests against known on-site inventory and mission requirements. 2. Produce a manifest optimiser compared against historical manifests on weight and shortfall. 3. Report weight used, requirement shortfall, redundant shipment rate. Baseline: historical manifest. Deliver Manifest planning dashboard.	Constraint satisfaction and knapsack-style optimisation; inventory reconciliation	Unit I search, CSP and optimisation; Unit II data cleaning; Unit V evaluation	Public expedition logistics and inventory records (Find and cite suitable public data yourselves.)	Add uncertainty in requirement estimates.	SMART INDIA HACKATHON 2026 PS ID: SIH26062 Integrated Polar Expedition Logistics and Asset Management System Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G30	Ayush Kumar Jayshwal (12511452) Giri Midhunsainathh (12500919)	**Sea-Ice and Iceberg Navigation Decision Support**	Vessels in ice-covered waters choose routes from static charts, while ice moves daily and the safe corridor shifts with it.	Polar research vessels, maritime operators	1. Build a decision support system that recommends a route given current ice conditions and reports the risk of each option. 2. Produce a route recommender evaluated against straight-line routing on transit risk. 3. Report transit time, cumulative ice risk, route feasibility. Baseline: shortest great-circle route. Deliver Chart interface with recommended route overlay.	Constrained routing over a changing cost field; trajectory prediction	Unit I search and constrained shortest path; Unit VI time-series; Unit V evaluation	Public sea-ice concentration products (NSIDC) (Find and cite suitable public data yourselves.)	Predict short-horizon ice drift and route against the forecast.	SMART INDIA HACKATHON 2026 PS ID: SIH26059 AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G31	Naman Kumar Nahar (12511086) Aniket Kumar Singh (12510117)	**Real-Time Inspection Scheduling and Monitoring App**	Surprise inspections are allocated ad hoc, so some institutions are visited repeatedly and others never.	Scheme monitoring bodies, inspectors, institutions	1. Build a scheduling system that allocates inspections by risk and coverage, with a monitoring dashboard. 2. Produce a risk-based inspection allocator compared against random allocation on issues found. 3. Report issues found per inspection, coverage spread. Baseline: random and round-robin allocation. Deliver Inspector mobile dashboard with assigned visits.	Risk scoring; allocation under coverage constraints; scheduling	Unit III classification for risk; Unit I optimisation; Unit V evaluation	Public institutional inspection records (Find and cite suitable public data yourselves.)	Add a fairness constraint guaranteeing minimum coverage.	SMART INDIA HACKATHON 2026 PS ID: SIH26095 Smart Real-Time Monitoring & Inspection Mobile App Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G32	Utkarsh Chauhan (12500852) Raman Rana (12512531)	**Safety Incident Precursor Detection from Field Reports**	Unsafe-act and unsafe-condition reports are filed and filed away, though they contain the precursors of the next serious injury.	Industrial HSE teams, field supervisors	1. Build an NLP engine that classifies free-text safety reports and surfaces precursor patterns before a serious incident. 2. Produce a report classifier plus a precursor trend report validated against recorded incidents. 3. Report per-class F1, precursor lead time, precision on flagged sites. Baseline: keyword rules. Deliver HSE dashboard with precursor alerts.	Text classification; topic discovery; trend detection over time	Unit V NLP and text representation; Unit VI clustering; Unit V evaluation	Public occupational safety incident narrative datasets (Find and cite suitable public data yourselves.)	Add site-level risk scoring from report mix.	SMART INDIA HACKATHON 2026 PS ID: SIH26165 AI/NLP Engine to Detect Serious Injury & Fatality (SIF) Precursors in OIL's Unsafe-Act/Unsafe-Condition and Near-Miss Reports Organisation: Oil India Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G33	Sahil Sarkar (12514482) Sujal Saklani (12512475)	**Infrastructure Project Progress Capture and Schedule Linking**	Site progress is reported in free text and photographs that never link back to the schedule, so slippage is invisible until milestones are missed.	Infrastructure project managers, contractors, funders	1. Build a system that extracts progress facts from site reports and links them to schedule activities, flagging divergence. 2. Produce an extraction and linking pipeline with linking accuracy and a divergence report. 3. Report extraction F1, linking accuracy, divergence detection lead time. Baseline: manual keyword tagging. Deliver Project dashboard with plan-versus-actual view.	Information extraction from text; entity linking to schedule items; divergence detection	Unit V NLP; Unit I knowledge representation; Unit V evaluation	Public construction progress report corpora (Find and cite suitable public data yourselves.)	Add photo-based progress evidence.	SMART INDIA HACKATHON 2026 PS ID: SIH26122 Intelligent Data Capture & Schedule-Linking Layer for Infrastructure Project Management: Real-Time Actual Progress Tracking (Planning-to-Execution Bridge) Organisation: Oil India Limited https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G34	Ranbeer Singh (12508980)	**Accelerated Optimisation Solver for Planning Problems**	Teams depend on commercial optimisation solvers, and there is no open benchmark of what an accessible solver can achieve on real planning problems.	Operations research teams, public sector planners	1. Build an implemented solver benchmarked against an open reference on standard planning and scheduling instances. 2. Produce a working solver benchmarked on standard instances against an open reference solver. 3. Report solution quality gap, time to solution, instances solved. Baseline: open reference solver, mandatory. Deliver Benchmark report with reproducible scripts.	Search and optimisation algorithms; metaheuristics; benchmarking methodology	Unit I search, heuristics and optimisation; Unit V benchmarking	Standard OR benchmark instance libraries (public) (Find and cite suitable public data yourselves.)	Add a parallel or accelerated variant and report speed-up.	SMART INDIA HACKATHON 2026 PS ID: SIH26119 Indigenous GPU-Accelerated Optimization Solver (Sovereign Alternative to Express / CEPLEX) Organisation: Mangalore Refinery and Petrochemicals Limited (MRPL) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data